1. For answers to 1a and 1b, please see the last PDF page.

2. **Adding Tiny Floating Point**

   a) My adding tiny floating point function covers a significant amount of "regular" functionality while breaking under some smaller equivalence classes of subcases.

   b) Here is everything I have tested to be functional:

      1. TFPs with a) same exponents, 2) positive, 3) normalized, 4) mantissa overflow and adjustment with rounding.
      2. TFPs with a 1) different exponents, 2) positive, 3) normalized, 4) mantissa overflow and adjustment with rounding.
      3. TFPs with 1) same exponent, 2) positive, 3) denormalized.
      4. TFPs with 1) mixed case (normalized and denormalized)
      5. TFPs with 1) same exponent, 2) same (negative) sign, 3) normalized
      6. TFPs with 1) same exponent, 2) different signs, 3) normalized
      7. TFPs with 1) different exponent, 2) different signs, 3) normalized

   c) Here is everything I *know* or *suspect* will break this function

      1. Special cases: special encoding, NaN, Infinity +/-, etc.
      2. Addition of mixed-sign denormalized numbers (see Case 12 in my test function)
      3. Cases where the addition of two TFPs results in an exponent that will be special-encoding/infinity

3. **Multiplying Tiny Floating Point**

   a) I was unfortunately not able to start this problem. I will continue working over spring break and turn this in for the re-dos assignment!

Problem 1: Floating Point Increments

1

a) Difference between largest and second largest non-infinite number?

largest: 0 1110 111 $= 1.111 \cdot 2^{14-7} = 1.111_2 \cdot 2^7$

s Exp Mantissa

2nd largest: 0 1110 110 $= 1.110 \cdot 2^{14-7} = 1.110_2 \cdot 2^7$

s Exp M

$= 0.001 \cdot 2^7$

$1.00 \cdot 2^4$

$\boxed{1.00_2 \cdot 2^4} = \boxed{16}$

b) Difference between 2nd smallest and smallest positive number

2nd Smallest: 0 . 0000 010 $= 0.010_2 \cdot 2^{1-7} = 0.010 \cdot 2^{-6}$

s exp m

Smallest: 0 0000 001 $= 0.001_2 \cdot 2^{1-7} = 0.001 \cdot 2^{-6}$

s exp m

$= 0.001 \cdot 2^{-6}$

$\boxed{1.00_2 \cdot 2^{-9}} = \boxed{\frac{1}{512}}$